

Affine source-reuse factor-three endpoint and global origin gap

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

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1. Purpose and scope: affine source reuse at the origin

R-154 proved a global origin-Hessian gap after requiring all source and target classes to be pairwise disjoint. R-155 removes exactly that restriction for the fixed-law affine innovation chart. Let $\mathcal{P}$ be any finite set of distinct nonzero retained dual-lattice momentum classes modulo sign. Retain every doubled class $[2p]$, but allow $\mathcal{P} \cap 2\mathcal{P} \neq \emptyset$. Define the zero-control Gaussian law once on the union $\mathcal{P} \cup 2\mathcal{P}$: one whitened raw innovation $\xi_{[r]}$ is attached to each antipodal class $[r]$. There is no duplicate Gaussian when a target is later reused as a source.

Order the roots predictably and use the R-128 fixed-law convention

$$h_{[2p]} = t_p H_p \xi_{[p]}, \quad H_p \in \mathbb{R}^{12 \times 12}. \quad (1.1)$$

The feedback functions in (1.1) are held fixed during the control variation; in particular, the already shifted state at $[p]$ is not inserted in place of the raw $\xi_{[p]}$.

At the centered stationary common-even covariance-matched A1/A6/A7 law, with fixed finite cut-off, positive floor, exact nonaliased torus integration, and the registered scalar regulator envelopes, the complete action Hessian obeys

$$D^2 \mathcal{A}(0)[H, H] > \frac{7}{250} \sum_{[p] \in \mathcal{P}} \|H_p\|_{\text{HS}}^2, \quad H \neq 0. \quad (1.2)$$

For the pure dyadic source-reuse family $\mathcal{P} \subset \{[2^j p_0] : j \in \mathbb{Z}\}$, with no extra vertices, the sharper bound is

$$D^2 \mathcal{A}(0)[H, H] > \frac{147}{1000} \sum_{[p] \in \mathcal{P}} \|H_p\|_{\text{HS}}^2. \quad (1.3)$$

The graph assertion below concerns the complete mixed covariance-normal endpoint/loss Hessian. The full Hessian is the diagonal Cameron–Martin form plus that endpoint Hessian plus one global positive-semidefinite sextic form. No sparsity is asserted for individual sextic mixed entries.

R-155 is an origin-only theorem for a canonical affine chart. It does not cover nonzero past, nonlinear or revisit feedback, finite-amplitude convexity, aliasing, an independent historical low coordinate, cutoff or floor removal, T-050 or A13 closure, Nelson, an interacting measure, any phase/PDE verdict, or Sector-A closure.

2. General source-target reuse algebra

The field and current along (1.1) are affine:

$$W_t = W_0 + \sum_p t_p S_{2p} H_p \xi_p, \quad V_{t,i} = V_{0,i} + \sum_p t_p D_{2p,i} H_p \xi_p. \quad (2.1)$$

Multiplication by two is injective on the torsion-free dual lattice, also after quotienting by antipodes:

$$[2p] = [2q] \implies 2p = \pm 2q \implies p = \pm q \implies [p] = [q]. \quad (2.2)$$

Thus distinct controls have distinct target blocks. More importantly, their raw source innovations are independent even when $[2p]$ is itself a member of $\mathcal{P}$. Pair blocks are not called independent: they share a class in that case. What remains independent are the distinct raw source directions entering (2.1).

For arbitrary $p, q \in \mathcal{P}$, whitened independence and (2.2) give the index identity

$$\mathbb{E}\langle H_p \xi_p, H_q \xi_q \rangle = \delta_{[p],[q]} \langle H_p, H_q \rangle_{\text{HS}}, \quad (2.3)$$

in the distinct doubled target blocks. Hence the complete source cost is

$$\mathcal{A}_{\text{CM}}(t) = \frac{9}{20} \sum_p t_p^2 \|H_p\|_{\text{HS}}^2, \quad D^2 \mathcal{A}_{\text{CM}}(0) = \frac{9}{10} \sum_p \|H_p\|_{\text{HS}}^2. \quad (2.4)$$

It is paid once. Source-target reuse creates no second payment and no off-diagonal source Gram entry.

Because (2.1) is affine and the raw sources are distinct, every mixed second covariance jet for $p \neq q$ vanishes:

$$C_{pq} = Q_{pq} = K_{pq,i} = 0. \quad (2.5)$$

This conclusion would fail for a chart whose second control reads a shifted state. That different chart is treated separately in Section 6.

3. Complete mixed endpoint and legal adjoint

Let $C = \text{Cov}(W)$, $Q = \sum_i \text{Cov}(V_i)$, and $K_i = \text{Cov}(W, V_i)$. The first jets are exactly the R-151 expressions

$$\begin{aligned} C_p &= S_{2p} H_p S_p^* + S_p H_p^* S_{2p}^*, \\ Q_p &= \sum_i (D_{2p,i} H_p D_{p,i}^* + D_{p,i} H_p^* D_{2p,i}^*), \\ K_{p,i} &= S_{2p} H_p D_{p,i}^* + S_p H_p^* D_{2p,i}^*. \end{aligned} \quad (3.1)$$

At $Q(0) = Q_{\text{A7}}$ and $K_i(0) = 0$, differentiation of the one A7 scalar gives, for two distinct controls,

$$\begin{aligned} D_{pq}^2 \mathcal{R}(0) &= \int \left\{ \frac{1}{2} \text{Tr}(Q_{pq} \overline{B}) \right. \\ &\quad + \frac{1}{4} [(Q_p)_{mn} (C_q)_{ab} + (Q_q)_{mn} (C_p)_{ab}] \overline{T}_{abmn} \\ &\quad \left. + \frac{1}{2} \sum_i [(K_{p,i})_{am} (K_{q,i})_{bn} + (K_{q,i})_{am} (K_{p,i})_{bn}] \overline{T}_{abmn} \right\} dx. \end{aligned} \quad (3.2)$$

The Q_{pq} term is displayed because it is required in a general curved or recursive chart. It vanishes by (2.5) in the canonical fixed-law chart.

Equation (3.2) contains both orientations already. The legal adjoint is $D_{qp}^2 \mathcal{R} = (D_{pq}^2 \mathcal{R})^*$; it is not a second reverse control, and it does not change the physical composition order. In particular, for $q = 2p$ both the forward block and its legal adjoint are zero by the Fourier calculation below.

4. Factor-two overlap is silent; factor-three remains

The support of every first response in (3.1) is derived from the target and source carriers:

$$\{2\epsilon - \epsilon' : \epsilon, \epsilon' \in \{-1, 1\}\}p = \{\pm p, \pm 3p\}. \quad (4.1)$$

For the reused source $q = 2p$, the second response has support $\{\pm 2p, \pm 6p\}$, disjoint from (4.1). Exact torus integration therefore gives

$$D_{p,2p}^2 \mathcal{R}(0) = 0. \quad (4.2)$$

For arbitrary distinct classes, cancellation between two copies of (4.1) leaves only

$$[q] = [3p] \quad \text{or} \quad [p] = [3q]. \quad (4.3)$$

Consequently the complete mixed endpoint/loss graph is again a union of finite factor-three paths. Source-target overlap by a factor two creates no new endpoint edge. This is not a claim that the complete origin Hessian is sparse: the terminal sextic Hessian remains one undecomposed global PSD form.

The complete mixed coefficient and R-130/R-151 envelopes are unchanged from R-154. The Q - C and K - K contributions at a factor-three edge are 240 and 324; antipodal normalization gives

$$4(240 + 324) = 2256. \quad (4.4)$$

Writing

$$f(x) = x^2 + Zx + r + \frac{7}{250}, \quad x = |p|^2, \quad (4.5)$$

the diagonal loss and one child-edge norm obey

$$L(x) = \frac{624(7083/2000)x}{4096f(x)f(4x)}, \quad E(x) = \frac{2256(7083/2000)x}{4096\sqrt{f(x)f(4x)f(9x)f(36x)}}. \quad (4.6)$$

5. Sharpened exact row certificates

Every nonzero lattice mode has $x > 3/20$. For a vertex without a factor-three parent, the sharpened target is

$$L(x) + E(x) < \frac{109}{125}. \quad (5.1)$$

Put $A = 624(7083/2000)/4096$, $D = 2256(7083/2000)/4096$, $F_{14} = f(x)f(4x)$, and $F_{936} = f(9x)f(36x)$. The exact sign guard is

$$\frac{109}{125}F_{14} - Ax > 0. \quad (5.2)$$

After (5.2), squaring (5.1) reduces to positivity of

$$\left\{ \frac{109}{125}F_{14} - Ax \right\}^2 F_{936} - D^2 x^2 F_{14}. \quad (5.3)$$

The polynomial in (5.3) has degree twelve. Its rational Sturm chain has length thirteen, six sign variations both at $3/20$ and at positive infinity, and a positive lower value. Thus it has no half-line root.

An independent standard-library implementation uses exact Fraction arithmetic, compactifies the half-line, and proves positivity by Bernstein coefficients on

$$[0, 1/64], [1/64, 1/32], [1/32, 1/16], [1/16, 1/8], [1/8, 1/4], [1/4, 1/2], [1/2, 1]. \quad (5.4)$$

The parent-row certificates imported from and rechecked against R-154 give

$$L(x) + E(x) + E(x/9) < \frac{1}{30} + \frac{1}{1000} + \frac{1}{6} = \frac{201}{1000}, \quad (5.5)$$

so that row has margin 699/1000 and is not the worst case. Combining (2.4), (5.1), and (5.5), and dropping the complete global sextic Hessian as one PSD form, proves (1.2):

$$\frac{9}{10} - \frac{109}{125} = \frac{7}{250}. \quad (5.6)$$

For a pure dyadic family, three is not a power of two; hence (4.3) produces no edge at all. A separate quartic Sturm certificate and independent Bernstein cover prove $L(x) < 753/1000$ for all $x \geq 0$, and therefore

$$\frac{9}{10} - \frac{753}{1000} = \frac{147}{1000}, \quad (5.7)$$

which is (1.3). This is an envelope certificate, not an assertion about an uncomputed exact tensor eigenvalue.

6. Separate controlled-state recursive companion

For clarity, R-155 also audits a different, explicitly declared chart:

$$\eta_{2p} = \xi_{2p} + tH\xi_p, \quad \eta_{4p} = \xi_{4p} + sG\eta_{2p}. \quad (6.1)$$

Then

$$W_{t,s} = W_0 + tS_{2p}H\xi_p + sS_{4p}G\xi_{2p} + tsS_{4p}GH\xi_p. \quad (6.2)$$

The bilinear chart connection is $z_{\text{conn}} = S_{4p}GH\xi_p$. On a single-parameter diagonal path the second derivative contains $2z_{\text{conn}}$; that factor two is not a second connection block and never reverses GH to HG .

The connection response has the derived support

$$\{4\epsilon - \epsilon' : \epsilon, \epsilon' \in \{-1, 1\}\}p = \{\pm 3p, \pm 5p\}. \quad (6.3)$$

The source cost is

$$\frac{9}{20} \left(t^2 \|H\|_{\text{HS}}^2 + s^2 \|G\|_{\text{HS}}^2 + t^2 s^2 \|GH\|_{\text{HS}}^2 \right), \quad (6.4)$$

so its origin Hessian remains diagonal. In (3.2) the connection contributes $\frac{1}{2} \int \text{Tr}(Q_{ts}\overline{B})dx$, but (6.3) makes it zero at this centered stationary covariance-matched, translation-invariant, common-even, exactly integrated zero-control origin. Gaussian integration by parts and the same support make the sextic first-variation connection zero there.

These vanishings are not a nonlinear-feedback theorem. Away from the declared origin, with nonzero past, covariance mismatch, aliasing, or a general revisit law, the connection terms must be retained and re-estimated.

7. Devil's-advocate review

1. **Objection: a reused class receives two independent Gaussians. DISMISSED.** The base law is defined once on $\mathcal{P} \cup 2\mathcal{P}$. Reuse changes incidence, not the raw variable.
2. **Objection: overlap destroys the source Gram. DISMISSED IN THE FIXED-LAW CHART.** Equation (2.3) uses distinct raw sources and injective doubled targets. It does not assert independence of whole overlapping pair blocks.
3. **Objection: factor two creates a hidden forward or reverse edge. DISMISSED.** The derived supports in (4.1)–(4.2) are disjoint. The only legal reverse is the adjoint of the same zero block.

4. **Objection: the factor-three graph describes every sextic mixed entry. UPHELD AS A WORDING CORRECTION.** The graph describes only the complete mixed endpoint/loss Hessian. The global sextic Hessian is kept as one PSD form and then dropped wholesale.
5. **Objection: a seven-root numerical chain proves an arbitrary finite theorem. DISMISSED AFTER REPAIR.** Both executables now use the generic two-index injection/independence algebra; carrier supports are derived, not assigned. The proof of the arbitrary-finite quantifier is the index calculation (2.2)–(2.5) plus the degree-two path row estimate.
6. **Objection: the recursive companion proves nonlinear or finite-amplitude reuse. UPHELD AS A BOUNDARY.** Section 6 is a separate two-control all-zero-origin audit. Its higher-order t^2s^2 source term is precisely where genuine reuse first appears beyond the Hessian.
7. **Objection: this favours BCC or validates the current PDE. UPHELD AS PROHIBITED.** R-155 is phase-neutral Hessian mathematics. It neither selects nor excludes BCC, a uniform condensate, another modulated state, metastability, or any effective equation.

8. Reproduction, route decision, and Result footer

From the repository root, run

```
E:\Dev\TECT.venv\Scripts\python.exe codes/foundations/
a13_classii_affine_source_reuse_factor_three_global_gap_boundary.py
E:\Dev\TECT.venv\Scripts\python.exe codes/foundations/
a13_classii_affine_source_reuse_factor_three_global_gap_boundary_independent.py
E:\Dev\TECT.venv\Scripts\python.exe codes/foundations/
a13_classii_affine_source_reuse_factor_three_global_gap_boundary_verify.py
```

R-155 is a real but deliberately bounded advance. It removes the affine origin theorem's source-target-disjointness hypothesis and shows that pure factor-two reuse is silent at second order. It does not make the direct T-050 route the sole research priority. The next direct A13 step, if taken, must instantiate nonzero past or genuinely nonlinear/revisit feedback from one scalar production action. In parallel, T-054 should freeze the phase-neutral evidence and candidate-equation comparison; T-052 may reject models early, and T-053 should be spent only on survivors. Truth, not BCC or any preselected PDE, is the selection criterion.

Result ID: A13-CLASSII-AFFINE-SOURCE-REUSE-FACTOR-THREE-GLOBAL-GAP-BOUNDARY (R-155).
Claim: A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION.
Precise statement: Affine source-reuse charts have gap $>7/250$; pure dyadic families have gap $>147/1000$, both in the summed HS source norm.
Scope: Fixed cutoff/floor, centered stationary covariance-matched common-even base, exact torus, distinct raw sources, affine origin Hessian.
Dependencies: A1, A7, R-128, R-130, R-151, R-154, and Sections 1--3.
Evidence grade: ANALYTIC, EXACT, EXECUTED, AUDITED.
Reproduction command: E:\Dev\TECT.venv\Scripts\python.exe codes/foundations/a13_classii_affine_source_reuse_factor_three_global_gap_boundary_verify.py
Expected output: Integrated 200/200 PASS; primary 49/49; independent 44/44.
Falsification gate: Any incidence, support, coefficient, exact certificate, PSD ownership, authority, executable, record, or PDF gate fails.
Tier before / after: T4 / T4; no promotion.
No-overclaim statement: No nonzero-past, general nonlinear/revisit, finite-amplitude, T-050/A13, Nelson/measure, phase/PDE, or Sector-A result follows.
Next required action: Minimum direct A13 successor plus T-054/T-052 in parallel.